

integration, environment-friendly machines for RoHS-based production, machines that can handle smaller components at a faster speed, and small yet intelligent machines.

With continuous evolution of electronics manufacturing processes, SMT equipment vendors are also upgrading their products and offering cost-effective high-technology solutions, while ensuring provision of value-adding services to their customers. Value-adding services are a major differentiator in the SMT market, while it is a major selection criterion for customers.

Speed and flexibility. SMT machines that simultaneously pick-and-place all components on a board are considered the fastest. But in this time and age, speed alone does not suffice; the electronics manufacturing processes need much more flexibility in order to enhance performance, reduce inventories and introduce new products. Hence, SMT machines that adopt an optimal balance of simultaneous and sequential operation, for example, simultaneous picking of components and sequential placement to enable flexibility and customisation, are available today. Machine manufacturers are also customising pick-and-place process in order to achieve the right mix of flexibility and speed. NMTronics' Fuji NXTII is a good example of flexibility. Its placement head is exchangeable to suit production requirements.

Scalability. Most top players provide modular SMT equipment, which gives you the flexibility to scale-up the SMT line as and when required. With Asmbleon's A series solution, you can simply hire additional ultra-compact robot heads to handle any temporary boosts in production with the same machine. "Agate & Agate's EM 760L can handle high-brightness and high-power LEDs, and can be configured to function as versatile multi-function SMT mounter. Even after attaching auto tray changer unit, this machine has 80 feeder slots of 8 mm,

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and in absence of auto tray changer unit it can take up to 120 feeders of 8 mm," informs Ashutosh Agate, managing director, Agate & Agate Marketing Resources.

Quick changeover. In this age of competition, to cut down on production cost, if a machine stays idle even for a second, money is lost. Therefore machines should ideally support offline setup and programming of feeders while the machines are still running, so that the changeover can be very swift. Also, to keep costs within control and to enable uninterrupted production, SMT equipment including placement tools and screen printers should ideally support high-speed

changeover.

There are two changeover strategies commonly used in machines—one is to retrofit the required changes alone, and the other is to treat the change as a completely new setup to ensure optimised position of the feeders. For example, Juki machines work on the concept of 3E—expandable, easy to use and economical. These machines are most flexible modular mounters, and the changeover takes few minutes due to trolley changeover system.

Accuracy. With electronic components getting smaller and smaller, SMT equipment are also becoming more capable of not just handling these components but also placing them accurately and close together, so that more components can be packed into a smaller space. The current trend is, component size capability of 0201 and 01005, and a mount precision of around 30 micron. Companies like Asmbleon have machines capable of 20 micron, operating with three-sigma accuracy. For its accurate mounting of bare dies and package-on-package, AX Platform is also suitable for semiconductor applications.

Superfine printing. An SMT printer is designed to provide accuracy needed for applying solder paste to PCBs for SMT production. Printing accuracy, speed and cost of ownership are the main challenges facing the screen printing market today. High-speed, automatic printers with 12-second cycle-time and 10-micron repeatable accuracies at fine pitch, represent the current technology. Specialised high- and low-volume printers are available in the market today. Essemtec AG's Tucano is a fully automatic inline printer, which optimally combines the requirements on quality and accuracy with speed and reliability. Due to its flexibility, Tucano can be adapted perfectly to virtually every print task. Also, customer-specific solutions can be realised within a short period of time.

Quality inspection. SMT pro-